Continuous Rearrangement and a Transport Picture for Quantitative Faber–Krahn / Saint–Venant Stability

Toward a selection-free proof

Abstract

We develop a geometric, non-perturbative route to the sharp quantitative Saint–Venant (equivalently Faber–Krahn) inequality that is intended to replace the Brasco–De Philippis–Velichkov (BDV) selection principle by a deterministic *continuous rearrangement flow*, in the spirit of the Dolbeault–Esteban–Figalli–Frank–Loss (DEFFL) proof of sharp Sobolev stability. The starting point is an exact splitting of the torsion deficit, $\delta_T = \mathcal{D}_{\rm rr} + \mathcal{R}_{\rm rad}$, into a *non-radiality* term $\mathcal{D}_{\rm rr}$ (the Pólya–Szegő gap of the torsion function) and a *radial-profile* term $\mathcal{R}_{\rm rad}$. The radial-profile term is controlled by a one-dimensional spectral gap and is elementary. The whole difficulty is concentrated in a single *quantitative Pólya–Szegő* estimate for $\mathcal{D}_{\rm rr}$, which we propose to prove by Brock's continuous Steiner symmetrization together with a competing-symmetries schedule, exactly as DEFFL do for Sobolev. We explain the transport interpretation (rearrangement as a measure-preserving deformation, dissipation as a transport cost), the monotone quantity along the flow, and why this route never touches the regularity of $\partial\Omega$ and hence bypasses selection. The remaining nucleus is stated as a precise open problem.

Contents

1 The transferable idea from DEFFL

The DEFFL proof of sharp stability for the Sobolev inequality [1] runs, schematically, as follows. Let $E(f)$ be the Sobolev deficit (the normalised gap in the inequality), whose zero set is the manifold $\mathcal{M}$ of Aubin–Talenti optimisers. They construct a deformation

$$f \rightsquigarrow f^* \quad \text{(symmetric decreasing rearrangement)},$$

implemented *continuously* through Brock's continuous Steiner symmetrization [2, 3], chained over a sequence of hyperplanes (a Bucur–Henrot schedule), and accelerated by *competing symmetries* (conformal rotations interleaved with rearrangement, after Carlen–Loss [4]). Two facts make this a proof of *quantitative* stability:

(i) *Monotonicity / dissipation.* The Dirichlet energy is non-increasing along the flow, and the rate at which it decreases is an explicit non-negative *dissipation* measuring how far the current profile is from being symmetric in the active direction.

(ii) *Local-to-global.* The flow drives any f to within a controlled distance of $\mathcal{M}$; one then only needs a *local* (near-optimiser) coercivity, supplied by the second variation. The deficit controls the total dissipation, hence the total flow distance, hence the distance of f to $\mathcal{M}$.

The conceptual upshot: **the continuous rearrangement flow is a deterministic substitute for a selection/compactness step.** It transports an arbitrary competitor to the optimiser while the deficit pays for the trip. No penalised minimisation, no subsequence, no contradiction.

The thesis of this note is that the *same* architecture should prove sharp quantitative Saint–Venant, with the torsion function u_Ω playing the role of f and the ball playing the role of the Aubin–Talenti manifold. The role of BDV's selection principle is taken over by the rearrangement flow.

2 An exact rearrangement splitting of the torsion deficit

Throughout, $\Omega \subset \mathbb{R}^n$ is bounded open, $|\Omega| = \omega_n$ (the volume of the unit ball $B = B_1$), and $u = u_\Omega \in H_0^1(\Omega)$ solves $-\Delta u = 1$ in Ω, $u = 0$ on $\partial\Omega$, with torsion $T(\Omega) = \int_\Omega u$. Write u^* for the symmetric decreasing rearrangement of u (extended by 0); $u^* \in H_0^1(B)$ and u^* is radial. Recall the convex/variational form

$$T(D) = \max_{v \in H_0^1(D)} J(v), \qquad J(v) := 2\int v - \int |\nabla v|^2,$$

attained at $v = u_D$, the torsion function of D, where $J(u_D) = \int u_D = T(D)$.

Proposition 1 (Exact deficit splitting)**.** *With the notation above,*

$$\delta_T(\Omega) \;=\; \underbrace{\Big(\|\nabla u\|_2^2 - \|\nabla u^*\|_2^2\Big)}_{=:\,\mathcal{D}_{\mathrm{rr}}(\Omega)\,\geq\,0} \;+\; \underbrace{\Big(T(B) - J(u^*)\Big)}_{=:\,\mathcal{R}_{\mathrm{rad}}(\Omega)\,\geq\,0}. \tag{1}$$

Here $\mathcal{D}_{\mathrm{rr}} \geq 0$ is the Pólya–Szegő gap of the torsion function (vanishing iff u is, up to translation, radial decreasing), and $\mathcal{R}_{\mathrm{rad}} \geq 0$ is the gap between the rearranged profile u^* and the true ball torsion function u_B (vanishing iff $u^* = u_B$).

Proof. Since $-\Delta u = 1$ and $u = 0$ on $\partial\Omega$, integration by parts gives $\|\nabla u\|_2^2 = \int u = T(\Omega)$. Rearrangement preserves the distribution function, so $\int u^* = \int u$, whence $J(u^*) = 2\int u - \|\nabla u^*\|_2^2$. Therefore

$$\mathcal{D}_{\mathrm{rr}} + \mathcal{R}_{\mathrm{rad}} = \big(\|\nabla u\|^2 - \|\nabla u^*\|^2\big) + \big(T(B) - 2\int u + \|\nabla u^*\|^2\big) = \|\nabla u\|^2 - 2\int u + T(B).$$

Using $\|\nabla u\|^2 = \int u = T(\Omega)$ this equals $T(\Omega) - 2T(\Omega) + T(B) = T(B) - T(\Omega) = \delta_T(\Omega)$. Non-negativity of $\mathcal{D}_{\mathrm{rr}}$ is Pólya–Szegő; non-negativity of $\mathcal{R}_{\mathrm{rad}}$ holds because $u^* \in H_0^1(B)$ is a competitor and $T(B) = \max_{H_0^1(B)} J = J(u_B)$. $\square$

Remark 2. The splitting (1) is the exact bookkeeping behind the classical Schwarz-symmetrization proof of Saint–Venant ($\delta_T \geq 0$). What is new is to read it as a sum of *two independently coercive* non-negative defects and to attack each by its own mechanism: $\mathcal{R}_{\mathrm{rad}}$ by a one-dimensional spectral gap (§3), $\mathcal{D}_{\mathrm{rr}}$ by a continuous-rearrangement quantitative Pólya–Szegő (§4). Crucially, neither mechanism uses any regularity of $\partial\Omega$: rearrangement is defined for arbitrary measurable competitors. This is the precise sense in which the route is selection-free.

Remark 3 (Eigenvalue version)**.** The same splitting holds verbatim for the first Dirichlet eigenvalue. With $\varphi = \varphi_\Omega$ the positive first eigenfunction, $\|\varphi\|_2 = 1$, and φ^* its symmetric decreasing rearrangement,

$$\lambda_1(\Omega) - \lambda_1(B) = \big(\|\nabla\varphi\|_2^2 - \|\nabla\varphi^*\|_2^2\big) + \big(R(\varphi^*) - \lambda_1(B)\big), \qquad R(\psi) := \tfrac{\|\nabla\psi\|^2}{\|\psi\|^2},$$

the first term the Pólya–Szegő gap of the eigenfunction, the second the radial Rayleigh residual. We work with torsion below because the convex duality is cleaner; everything transfers.

3 The radial-profile term is elementary

$\mathcal{R}_{\mathrm{rad}} = J(u_B) - J(u^*)$ compares two radial functions on the fixed ball B. Since J is strictly concave with maximiser u_B, and the second variation of J at u_B is $-2\int|\nabla h|^2$, restricted to radial $h \in H_0^1(B)$, we get a clean one-dimensional coercivity.

Lemma 4 (Radial gap)**.** *There is $c_n > 0$ such that*

$$\mathcal{R}_{\mathrm{rad}}(\Omega) \;\geq\; c_n\, \|u^* - u_B\|^2_{H_0^1(B)}\,.$$

Proof. Write $u^* = u_B + h$ with $h \in H_0^1(B)$ radial and $\int(u^* - u_B)\,(-\Delta u_B - 1) = 0$ (the Euler–Lagrange orthogonality, automatic since $-\Delta u_B = 1$). Then $J(u^*) = J(u_B) - \int|\nabla h|^2$, so $\mathcal{R}_{\mathrm{rad}} = \int|\nabla h|^2 = \|\nabla h\|_2^2$, and $\|h\|^2_{H_0^1} = \|\nabla h\|_2^2$ by Poincaré on B. Hence $\mathcal{R}_{\mathrm{rad}} = \|\nabla h\|_2^2 \geq c_n\,\|h\|^2_{H_0^1}$ with c_n the Poincaré constant of B. ☐

In particular, $\mathcal{R}_{\mathrm{rad}}$ controls the deviation of the *radial profile* of u_Ω from the exact ball profile. This is the easy half of the deficit and needs nothing beyond a Poincaré inequality on the ball.

4 The non-radiality term and the continuous flow

The entire content of sharp stability is now in $\mathcal{D}_{\mathrm{rr}}$, the Pólya–Szegő gap. We must show it controls the distance of u_Ω from the radial cone, in a norm strong enough to recover the Fraenkel asymmetry. This is exactly where DEFFL's continuous machinery enters.

4.1 Brock's continuous Steiner symmetrization

For a direction (hyperplane) H, Brock's flow $\tau \mapsto u_\tau^H$, $\tau \in [0,\infty]$, continuously rearranges u along lines $\perp H$, interpolating $u_0^H = u$ and $u_\infty^H = u^{*H}$ (Steiner symmetrization in H). It preserves all L^p norms and the distribution function, and the Dirichlet energy is non-increasing,

$$\frac{d}{d\tau}\,\|\nabla u_\tau^H\|_2^2 \;\leq\; 0, \tag{2}$$

right-continuously in τ [2, 3]. Chaining a sequence of hyperplanes (H_k) with the Bucur–Henrot reparametrisation produces a single flow $\tau \mapsto u_\tau$, $\tau \in [0,\infty]$, with $u_0 = u$, $u_\infty = u^*$, $\|u_\tau\|_p = \|u\|_p$, and $\tau \mapsto \|\nabla u_\tau\|_2^2$ non-increasing. Thus

$$\mathcal{D}_{\mathrm{rr}}(\Omega) \;=\; \|\nabla u\|_2^2 - \|\nabla u^*\|_2^2 \;=\; \int_0^\infty \Big(-\tfrac{d}{d\tau}\,\|\nabla u_\tau\|_2^2\Big)\,d\tau \;=\; \int_0^\infty \mathfrak{d}(u_\tau)\,d\tau, \tag{3}$$

where $\mathfrak{d}(u_\tau) \geq 0$ is the instantaneous *dissipation*. The flow is the deterministic "transport to the ball": u is carried to its radial rearrangement, and $\mathcal{D}_{\mathrm{rr}}$ is the total energy released.

4.2 The monotone quantity and the local-to-global principle

The monotone quantity along the flow is the Dirichlet energy $\|\nabla u_\tau\|_2^2$ (equivalently the running torsion functional $J(u_\tau) = 2\int u - \|\nabla u_\tau\|^2$, non-decreasing). The DEFFL local-to-global principle, transported to our setting, reads:

> *If at no time τ the profile u_τ is close to radial, then the dissipation $\mathfrak{d}(u_\tau)$ stays bounded below, so $\mathcal{D}_{\mathrm{rr}} = \int \mathfrak{d}\, d\tau$ is large. Contrapositively, small $\mathcal{D}_{\mathrm{rr}}$ forces u_τ to be close to radial for a substantial range of τ, in particular near $\tau = 0$, i.e. $u = u_\Omega$ itself is close to radial.*

To make this quantitative one needs a lower bound on the dissipation by a *distance to the radial cone*. Denote by

$$\mathcal{C}_{\mathrm{rad}} := \{\, w^*(\,\cdot - z\,) : z \in \mathbb{R}^n \,\}$$

the cone of translated radial-decreasing rearrangements, and let $\operatorname{dist}(u, \mathcal{C}_{\mathrm{rad}})$ be measured in a norm to be specified. The nucleus of the whole program is:

Problem 5 (Quantitative Pólya–Szegő for the torsion function)**.** Find a norm $\|\cdot\|_X$ controlling the Fraenkel asymmetry of the super-level sets and a constant $c = c(n) > 0$ such that

$$\mathcal{D}_{\mathrm{rr}}(\Omega) \;=\; \|\nabla u\|_2^2 - \|\nabla u^*\|_2^2 \;\geq\; c \inf_{z\in\mathbb{R}^n} \|\, u - u^*(\,\cdot - z\,)\,\|_X^2 \,. \tag{4}$$

Two features of (4) must be respected, and they are exactly the features DEFFL engineer for Sobolev.

(a) Degeneracy of one-shot Pólya–Szegő. The bare gap $\|\nabla u\|^2 - \|\nabla u^*\|^2$ is *not* coercive in general: Cianchi–Fusco exhibit families where the gap is much smaller than the asymmetry, because a single rearrangement can be "almost free" while still moving mass. This is the rearrangement analogue of the obstruction we met elsewhere in this project. The cure, after DEFFL, is to not use one rearrangement but the *continuous flow*: integrate the dissipation (3) and bound it below using the spectral structure of the flow at *every* time, not just the endpoints.

(b) Competing symmetries to prevent stalling. A single Steiner direction can leave the profile invariant (if u is already symmetric in that direction) without being radial. DEFFL interleave conformal rotations so that the schedule cannot stall away from the optimiser. The Faber–Krahn analogue of the conformal group is absent (the torsion problem is not conformally invariant), so a *different* anti-stalling device is needed: the natural candidate is to randomise/cycle the Steiner directions densely (a.e. schedule converges, by [5]) and to use the *rotational averaging* of the dissipation. Establishing a uniform lower bound on the time-averaged dissipation in terms of non-radiality is the technical heart.

4.3 The instantaneous dissipation, explicitly

For a single Steiner direction $H = e^\perp$, slicing $\mathbb{R}^n = \mathbb{R}_e \times H$ and writing $u(s, y)$ ($s \in \mathbb{R}$, $y \in H$), Brock's dissipation has the one-dimensional-slice form

$$\mathfrak{d}_H(u) \;=\; \int_H \mathfrak{d}_1\big(u(\cdot, y)\big)\, dy, \tag{5}$$

where $\mathfrak{d}_1(g) \geq 0$ measures, for the one-dimensional profile g, the rate of Dirichlet-energy loss under the 1D continuous symmetrization. For a smooth slice with super-level intervals $\{g >$

$t\} = \bigcup_j (a_j(t), b_j(t))$, $\mathfrak{d}_1$ is a positive quadratic form in the velocities of the interval endpoints, vanishing exactly when the intervals are concentric. The key elementary fact is that $\mathfrak{d}_1(g)$ controls the *1D asymmetry* of g: how far the super-level sets of g are from being centred intervals. Integrating over slices $y \in H$ and over directions $H \in \mathbb{S}^{n-1}$, the dissipation controls the full (non-radial) asymmetry. Turning this chain of "controls" into a quantitative inequality with a uniform constant is Problem 5.

5 Transport interpretation

The rearrangement $u \mapsto u^*$ is a measure-preserving transport at the level of super-level sets: for each height t, the set $\{u > t\}$ is transported to the ball $\{u^* > t\} = B_{r(t)}$ of equal volume. The continuous flow u_τ realises this transport through a one-parameter family of measure-preserving maps T_τ, with $T_0 = \mathrm{id}$ and $(T_\infty)_\#$ sending each level set to its centred ball. In this picture:

- the *monotone quantity* is the Dirichlet energy $\int |\nabla u_\tau|^2$, decreasing along transport;
- the *dissipation* $\mathfrak{d}(u_\tau)$ is the instantaneous transport cost (the squared transport velocity weighted by the gradient);
- the *total cost* $\int_0^\infty \mathfrak{d}\, d\tau = \mathcal{D}_{\mathrm{rr}}$ is the deficit's payment for moving u to the ball.

This is the precise mirror of the isoperimetric "transport to the ball" heuristic, now at the level of the torsion function rather than the bare domain. The advantage over a raw domain transport is that the gradient weight in the cost makes spectrally inefficient features (thin tentacles, spikes, high-frequency boundary oscillation) *cheap to remove and yet large in energy*, which is exactly the favourable imbalance we verified directly via the Dirichlet-energy scaling of thin features in the companion energy-identity analysis.

Remark 6 (Relation to the pressure flow)**.** The companion note develops a *domain* flow with normal velocity $V_n = (\partial_\nu u)^2 - \fint(\partial_\nu u)^2$ (boundary-pressure equalisation), whose equilibrium is the ball by Serrin. That flow is intrinsically perturbative: it requires a smooth moving boundary, and is the infinitesimal/Hadamard picture. The rearrangement flow of this note is its *non-perturbative* counterpart: it acts on the function, not the boundary, is defined for arbitrary rough Ω, and integrates a global dissipation. The two are linked by the coarea formula — the pressure $(\partial_\nu u)^2$ is the boundary trace of $|\nabla u|^2$, whose super-level redistribution is what the rearrangement flow performs — but only the rearrangement flow survives at the rough, global scale where selection was previously needed.

6 Recovering the Fraenkel asymmetry

Granting Problem 5 in a norm X controlling super-level asymmetry, the deficit controls the domain asymmetry as follows. By the triangle inequality and Proposition 1,

$$\inf_z \|u - u_B(\,\cdot - z\,)\|_X \;\le\; \inf_z \|u - u^*(\,\cdot - z\,)\|_X + \|u^* - u_B\|_X \;\lesssim\; \mathcal{D}_{\mathrm{rr}}^{1/2} + \mathcal{R}_{\mathrm{rad}}^{1/2} \;\lesssim\; \delta_T^{1/2}.$$

The super-level sets of u are then close to those of $u_B(\cdot - z)$, uniformly in height. Two points remain, both standard in flavour but requiring care at the rough scale:

(i) *From level sets to the domain.* The Fraenkel asymmetry is the $t \to 0$ level $\{u > 0\} = \Omega$. Control of all positive levels plus the Hopf non-degeneracy $u \sim \mathrm{dist}(\cdot, \partial\Omega)$ near the boundary upgrades level-set control to $\mathcal{A}_F(\Omega) \lesssim \delta_T^{1/2}$. Unlike the single-good-level approach, the rearrangement controls *all* levels simultaneously, which should make the $t \to 0$ passage more robust.

(ii) *Sharp exponent.* The target is $\mathcal{A}_F(\Omega)^2 \le C\,\delta_T(\Omega)$, i.e. exponent 2. The splitting and Lemma 4 are already at the sharp quadratic scale; the exponent in the final estimate is inherited entirely from the exponent in (4). Thus *sharpness of the quantitative Pólya–Szegő is equivalent to sharpness of Faber–Krahn* along this route — a clean reduction.

7 Why this bypasses selection

(1) *No regularity of $\partial\Omega$ is ever used.* Rearrangement and Brock's flow are defined for arbitrary measurable u; the dissipation is a function-space quantity. The selection step existed only to manufacture a smooth near-equilibrium boundary on which BNST / second variation could run; here there is no boundary analysis at all until the very end (§6(i)), and even there only Hopf non-degeneracy is needed.

(2) *No compactness or contradiction.* The flow is explicit and deterministic, and $\delta_T = \mathcal{D}_{\mathrm{rr}} + \mathcal{R}_{\mathrm{rad}}$ is an identity. Stability is a *computation* of how the deficit is distributed between non-radiality and profile, exactly as DEFFL turn the Sobolev deficit into an explicit flow budget.

(3) *Global and non-perturbative.* The estimate (4), if true with a uniform constant, holds for all Ω, not just near the ball. The perturbative second-variation analysis (our (G), BDV Thm 3.3) is recovered as the *linearisation* of (4) at $u^* = u_B$, but is no longer the load-bearing input.

8 The nucleus, and an honest assessment

The program reduces sharp quantitative Saint–Venant, with no selection, to:

> **Problem 5, sharp form.** Prove $\|\nabla u\|_2^2 - \|\nabla u^*\|_2^2 \ge c(n)\inf_z \big\| \{u > \cdot\} \triangle \{u^*(\cdot - z) > \cdot\} \big\|^2$ in a level-set asymmetry norm at the quadratic scale, by integrating Brock's dissipation (3) along a non-stalling Steiner schedule.

What is solid:

- the exact splitting $\delta_T = \mathcal{D}_{\mathrm{rr}} + \mathcal{R}_{\mathrm{rad}}$ (Proposition 1), both terms non-negative, no regularity used;
- the radial half $\mathcal{R}_{\mathrm{rad}} \gtrsim \|u^* - u_B\|_{H^1}^2$ (Lemma 4), elementary;
- the reduction of the *exponent* to that of the quantitative Pólya–Szegő;
- the structural match with DEFFL: monotone Dirichlet energy, integrated dissipation, local-to-global via the flow.

What is open, and genuinely hard:

- Problem 5 itself. One-shot Pólya–Szegő is known to degenerate (Cianchi–Fusco); the continuous flow is the proposed cure but the dissipation lower bound (5) must be made quantitative and uniform.
- The *anti-stalling* device. The torsion problem lacks the conformal symmetry that DEFFL exploit via competing symmetries; a substitute (dense/random Steiner schedule with a time-averaged dissipation bound) must be supplied. This is the single place where the Sobolev proof does not transfer verbatim, and where new ideas are required.
- The $t \to 0$ passage from level-set control to domain asymmetry at the rough scale (§6(i)).

Verdict. The rearrangement/transport route gives a genuinely non-perturbative architecture for quantitative Faber–Krahn in which the selection principle is replaced by a deterministic flow and the entire difficulty is isolated in one quantitative Pólya–Szegő inequality. It is the most plausible candidate among the routes considered in this project for a proof with real geometric content that does not pass through BDV selection. Its viability hinges on Problem 5 and on finding the right anti-stalling substitute for competing symmetries; both are concrete and, in the Sobolev case, were solved exactly by DEFFL — which is the strongest available evidence that the torsion case is within reach.

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